THE HONG KONG POLYTECHNIC UNIVERSITY

BEng (Hons) in ME

Individual Interim Report

Design and Development of a Passive Vanadium Redox Flow Battery-Powered Autonomous Toy Car

By

ZHAO Xuecen (22014303D)

Chief Supervisor: Prof. AN Linag
Second Supervisor: Prof. WU Maochun

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1.INTRODUCTION

1.1 Global Energy and Environmental Challenges

The rise in global population has triggered a surge in energy demand, especially in transportation. Currently, transportations rely heavily on fossil fuels, which contribute to greenhouse gas emissions and global warming.
To reduce emissions, we need to find alternative to internal combustion engine (ICE) fuel. Renewable energy sources like wind and solar but they can only generate energy and cannot act as energy storage.

1.2 Limitations of Current Electric Vehicle Technology

Currently electric vehicles (EVs) powered by lithium-ion batteries is the primary alternative to fossil fuel vehicles. However, there are several limitations that slows down the adoption of EVs. Traditional charging methods are time-consuming. Lithium-ion batteries also suffer from degradation and have limited life cycles. There is also concerns about the safety risk of EVs, including battery overheating and ignition upon impact. These challenges make many users hesitant to swap their ICE vehicles to EVs, we need an alternative solution that combines the low-emission benefits of EVs with the rapid refuelling and safe to drive.

1.3 The Vanadium Fuel Cell Solution

To address these limitations, this project explores the application of Vanadium Redox Flow Battery (VRFB) adapted into a passive direct liquid fuel cell stack. Unlike conventional batteries, vanadium fuel cells have a liquid electrolyte that allows for rapid refuelling. Furthermore, vanadium electrolytes are non-flammable, offering safety advantages over lithium battery. The system also has long cycle life and high recyclability, making it a sustainable candidate for future mobility solutions.

1.4 Project Overview: The "Vandi" Autonomous Vehicle

The objective of this project is to design, and fabricate a electric toy car, named "Vandi," powered by passive vanadium fuel cell stack.
This interim report covers the initial design and fabrication phases:
Power System: Design and prototyped a vanadium fuel cell stack with modular design.
Mechanical Integration: Assembly of a mecanum-wheel chassis designed to carry the fuel cell stack, control system, and sensor.
Intelligent Control: AI navigation systems, depth cameras utilizing YOLO (You Only Look Once) algorithms for object detection, alongside a prototype for a Large Language Model (LLM) interface for user interaction.

1.5 Report Structure

This report records the progress during the first semester of the project.
Section 2 the battery technologies, vanadium fuel cell principles, and autonomous navigation strategies.
Section 3 the mechanical design, fuel cell stack prototype, and AI control systems.
Section 4 the implementation details, including the car body assembly, electrolyte optimization experiments, and YOLO model training results.
Section 5 the future work required to finalize the "Vandi" prototype.

Section 6 the project schedule and timeline.
Section 7 the necessary resources and budget.
Section 8 concludes the report.

2.Background and Literature Review

2.1 Overview of Fuel Cell Technologies

Fuel cells convert chemical energy directly into electricity. They are classified by their electrolyte and operating temperature.

Fuel Cell Type	Operating Temperature	Advantages	Disadvantages
Proton Exchange Membrane Fuel Cells (PEMFCs)	60–80°C (low temperature)	High-power density, quick start-up	Expensive platinum catalysts, requires pure hydrogen fuel
Direct Liquid Fuel Cells (DLFCs)	Not specified	Uses liquid fuels (e.g., methanol), supports portable applications	Fuel crossover, slow reaction kinetics, methanol toxicity
Solid Oxide Fuel Cells (SOFCs)	600–1000°C (high temperature)	Can use flexible fuels	Brittleness, long start-up times, limited use in mobile applications

2.2 Vanadium Redox Flow Batteries (VRFB)

Vanadium Redox Flow Batteries (VRFBs) is great for large-scale energy storage. A VRFB uses vanadium ions in four different oxidation states V^{2+}, V^{3+}at the negative electrode and VO^{2+}, $VO_2{}^{+}$at the positive electrode.

Ion Formula	Oxidation State	Role in VRFB	Colour
V^{2+}	+2	Negative electrolyte (charged)	Purple
V^{3+}	+3	Negative electrolyte (discharged)	Green
VO^{2+}	+4	Positive electrolyte (discharged)	Blue
VO_2^{+}	+5	Positive electrolyte (charged)	Yellow

Positive Electrode (Cathode) Reaction:
$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$
Negative Electrode (Anode) Reaction:
$V^{2+} \rightleftharpoons V^{3+} + e^-$
Overall Cell Reaction:
$VO_2^+ + V^{2+} + 2H^+ \rightleftharpoons VO^{2+} + V^{3+} + H_2O$

The key advantage of VRFBs:

Power is determined by the stack size, and energy capacity is determined by the volume of electrolyte stored in fuel tanks and the activity of electrodes.
This allows VRFBs to have easy scalability and long cycle life (>10,000 cycles).
However, traditional VRFBs requires pumps and flow management, which make it hard to apply into small devices.

2.3 The "E-Fuel" Concept and Passive Fuel Cells

Recent innovations have allowed VRFB technology for portable use. Shi et al. introduced the concept of "electrically rechargeable liquid fuel" (e-fuel), where the charged vanadium electrolyte V^{2+} serves as a liquid fuel that can refill instantly.
Using this for small-scale design, researchers developed passive direct liquid fuel cells. Unlike active systems, passive cells do not need pumps and blowers. Instead, it relies on diffusion and natural convection.

2.4 Autonomous Navigation in Small-Scale Vehicles

To demonstrate a small-scale smart vehicle, we decide to include intelligent navigation. Modern autonomous navigation relies on object detection models like YOLO that collaborate with an algorithm for path planning.
YOLO (You Only Look Once): The YOLO algorithm is a real-time object detection model with high speed and accuracy. For devices like small toy cars, there is also a lightweight version such as YOLOv11-Nano. These models can process inputs from cameras with high inference speeds, suitable for computing devices like the Jetson Nano.
Sensor: Effective navigation requires collecting data from depth cameras and LiDAR. This combination allows the vehicle to map its environment (SLAM) and execute path planning.

3.Design and Working Principle

3.1 Vehicle Mechanics

3.1.1 Chassis and Drive System

We purchased a car wheelbase equipped with DC motor, UART (Universal Asynchronous Receiver-Transmitter) wheels and chassis. The manufacturer provided us with CAD drawing and dimensions for custom modification.

The propulsion is provided by four 12V DC brushless motors equipped with magnetic encoders for closed-loop speed control. The motors drive 60mm diameter mecanum wheels. The relationship between the motor RPM (N) and the linear velocity of the wheel (v) is given by Equation (1):

$$v = \frac{\pi \cdot D \cdot N}{60} \quad (1)$$

Where D is the wheel diameter (0.06 m).

3.1.2 Kinematics

Kinematic modelling from ME31002 Linear Systems and Control was applied to derive the kinematic relationship of a four-wheel mecanum drive, relating the platform velocity to individual wheel angular speeds. Let v_x and v_y be the linear velocities in the longitudinal and

transverse directions, respectively, and be the angular velocity ω_z around the vertical axis. The angular velocity of each wheel ω_1, ω_2, ω_3, ω_4) is given by Equation (2):

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} 1 & -1 & -(l_x + l_y) \\ 1 & 1 & (l_x + l_y) \\ 1 & 1 & -(l_x + l_y) \\ 1 & -1 & (l_x + l_y) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix} \quad (2)$$

Where:

- r is the radius of the wheel.
- l_x is the half-distance between the front and rear wheels (wheelbase/2).
- l_y is the half-distance between the left and right wheels (trackwidth/2).

3.2 Fuel Cell Stack Design

The core power unit is a custom-designed passive direct liquid fuel cell stack utilizing vanadium-based fuel. Unlike active systems, this stack relies on natural convection and vibration, eliminating the need for pumps.

3.2.1 Electrochemical Reactions

The power subsystem is the main focus of this project because it determines the runtime and payload. Utilizing a passive vanadium-based direct liquid fuel cell (DLFC) stack to provide sustainable electric power.

Fuel Performance Analysis

The performance of the fabricated stack is evaluated by measuring its current and power density.

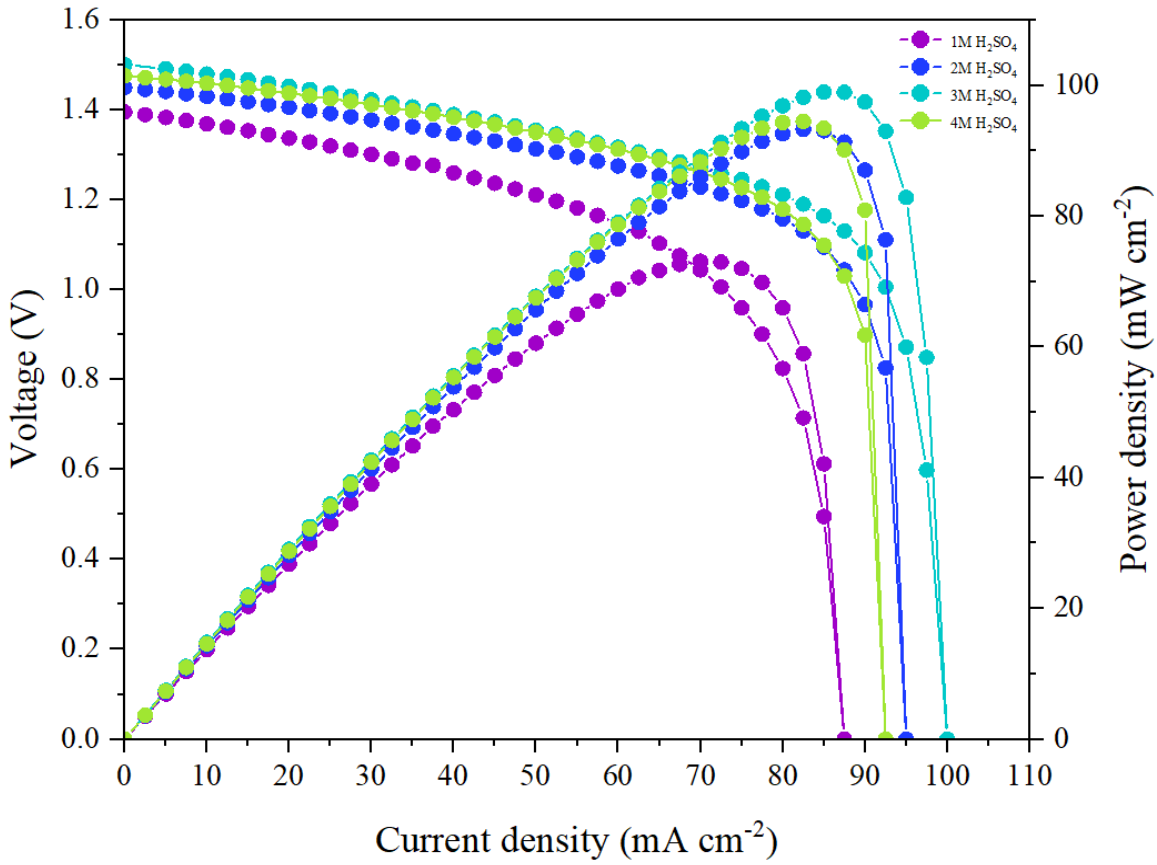

Figure 1 Performance analysis graph for different Sulfuric Acid concentration.

Anode Reaction (Oxidation): Vanadium(II) ions are oxidized to Vanadium(III), releasing an electron.

$$V^{2+} \rightarrow V^{3+} + e^{-}(E^{0} = -0.26\text{ V})(4)$$

Cathode Reaction (Reduction): Vanadium(VO_2^+) ions combine with protons and electrons to form Vanadium(VO^{2+}).

$$VO_2^{+} + 2H^{+} + e^{-} \rightarrow VO^{2+} + H_2O(E^{0} = +1.00\text{ V})(5)$$

Overall Cell Reaction:

$$VO_2^{+} + V^{2+} + 2H^{+} \rightarrow VO^{2+} + V^{3+} + H_2O(E^{0}_{cell} = 1.26\text{ V})(6)$$

The theoretical open-circuit voltage (OCV) for a single cell is 1.26 V. However, the actual operating voltage (V_{cell}) is lower due to activation losses η_{act}, ohmic losses η_{ohm}, and concentration losses η_{conc}, as described by Equation (7):

$$V_{cell} = E^{0}_{cell} - \eta_{act} - \eta_{ohm} - \eta_{conc}(7)$$

3.2.2 Stack Design

Fuel Cell Design (Single Cell)

In the ME44002 Engine Technology course, we were introduced to the reaction theory of fuel cells.

- Active Area: A geometric active $area(A_{cell})\ of$ 4.0 cm (2.0 cm × 2.0 cm) was selected to balance power output with fuel consumption rates.
- Current Collectors: Graphite plates or gold-plated copper plates
- Membrane Electrode Assembly (MEA):
    - Membrane: Nafion 212
    - Electrodes: Graphite felt.

Stack Configuration:
The final stack will consist of 6 single cells connected in series. Providing a theoretical open-circuit voltage (OCV) of 7.56 V.

3.3 AI Integration and Control Architecture

3.3.1 Hardware Architecture

Main Computer: NVIDIA Jetson Nano (8GB) running ROS (Robot Operating System) Melodic.
Low-Level Controller: Custom controller handling PWM generation for motor drivers and PID speed control.
Sensors:

- **2D LiDAR (RPLIDAR A1):** Provides 360-degree distance data for Simultaneous Localization and Mapping (SLAM).
- **RGB-D Camera:** Captures visual data for object recognition.

High-Level Processor (Brain):
We were first introduced to the NVIDIA Jetson embedded AI platform in ME39003 Multidisciplinary Manufacturing Project, where we learned how to operate the Jetson environment and develop programs directly on the system.

- **Device:** NVIDIA Jetson Nano Developer Kit (8GB RAM).
- **Role:** Runs the ROS (Robot Operating System) master node, handles image processing (YOLOv11), and performs SLAM calculations.
- **Mounting:** Placed on the elevated upper deck to ensure the camera has a clear field of view and the LiDAR is unobstructed by vehicle components.

Low-Level Controller (Body):
Device: Custom Microcontroller.
Role:

- Reads encoder data to calculate current wheel speed.
- Implements a PID (Proportional-Integral-Derivative) control loop to match actual wheel speed to the target velocity received from the Jetson Nano.
- Generates PWM (Pulse Width Modulation) signals for the motor drivers.

The PID control equation used for motor speed regulation is described as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \quad (9)$$

Where:

- u(t) is the control output (PWM duty cycle).
- e(t) is the error (Target Speed - Actual Speed).
- K_p, K_i, K_d are the proportional, integral, and derivative gain coefficients, tuned experimentally.

Basic movements via controller input
Manual driving is implemented as a controller-to-motor pipeline, allowing basic actions (forward, reverse, strafe, rotate, stop) to validate chassis motion before autonomous driving is enabled.
Sensor Integration:

- **LiDAR (RPLIDAR A1):** Mounted at the highest point of the vehicle (approx. 20cm from ground) to provide a 360° scan of the environment for obstacle avoidance.
- **Depth Camera:** Mounted at the front chassis bumper, angled slightly upwards to detect traffic signs and traffic lights.

3.3.2 Object Detection Algorithm (YOLO)

Real-time object detection is achieved using YOLO (You Only Look Once), learned in ME41600 Perceptual Robotics, enabling the vehicle to identify traffic signals and road features. Visual detection for this project is implemented using a YOLO-based object detection model, with the project target of achieving detection confidence above 80% for the relevant objects (road signs / traffic elements used on the demo track). Training data is collected from the demonstration environment (track images and traffic signal/sign samples), and the trained model is validated on held-out test data before deployment.

To detect traffic signs and obstacles, we utilize the YOLOv11 (You Only Look Once) neural network architecture. YOLO treats object detection as a single regression problem, predicting bounding boxes (B) and class probabilities (C) directly from full images in one evaluation. The loss function (L) for the model combines localization loss, classification loss, and confidence loss:

$$L = \lambda_{coord} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{obj} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2\right] + \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{obj} \left(\sqrt{w_i} - \sqrt{\hat{w}_i}\right)^2 + \cdots \quad (10)$$

The YOLO loss function combines localization, classification, and confidence loss terms. For this project, the training was performed using the ultralytics framework, which automatically computes these losses during the training process.

For this project, a lightweight version, **YOLO v11-Nano**, was selected to ensure real-time inference (>15 FPS) on the edge computing hardware, balancing accuracy with power consumption. The model is trained on a custom dataset of traffic signs (Stop, Go, Left, Right) to enable autonomous decision-making.

3.3.3 Interactive control through an external frontend

An external user-facing frontend is implemented as an interactive chat interface that supports command execution and Q&A during demonstration

4. System Implementation and Testing

This section describes the individual subsystems fabricated and tested, including mechanical assembly, fuel cell optimization, AI vision systems, and remote-control implementation.

Mechanical part:

car selection, assembly, are done together as a team. I am also fully responsible of car tray design.

Fuel testing:
Multiple lab sessions were conducted to understand the charge/discharge behavior of the vanadium electrolyte.

Fuel cell design:
I am fully in charge of the iteration and design of the fuel cell, while fuel cell testing was done in collaboration with BME teammate.

YOLO sample data collection:
Done in IC lab as a team.

AI training, remote control programming, and interactive interface:
YOLO training and interactive interface were performed individually.

4.1 Car Body Assembly

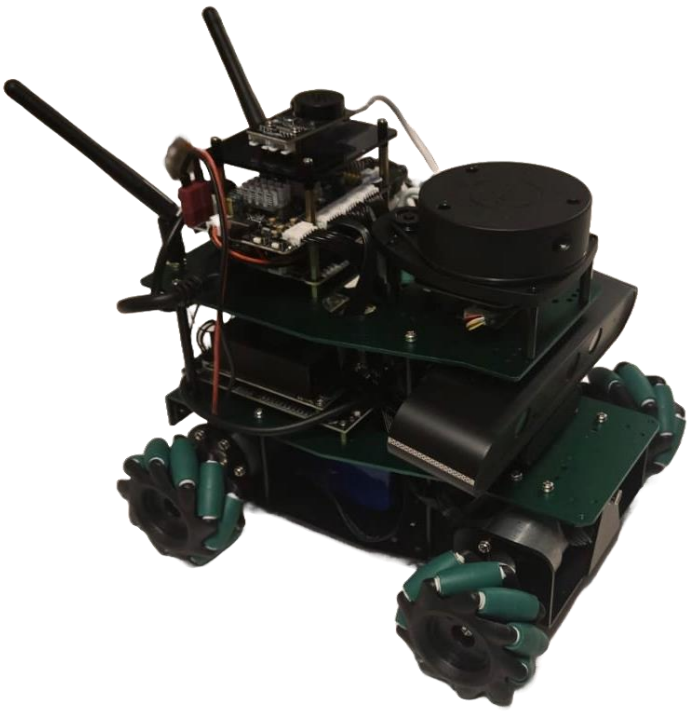

Figure 2 Fully assembled Vandi.

The physical construction of the autonomous vehicle, named "Vandi," has two primary subsystems: the mechanical drive base and the electronic control integration. The assembly process aimed to create a stable platform capable and enable further integration of a tray for carrying the fuel cell stack while maintaining agility for the demonstration track.

4.1.1 The drive base

The drive base is built upon a purchased chassis kit modified to accommodate our specific power and control requirements.

- **Chassis Structure:** The main body utilizes a rigid sheet metal. The chassis dimensions (approximately 23 cm × 15 cm).
- **Motion System:** To enable omnidirectional movement, the car is equipped with four mecanum wheels. Each wheel is independently driven by a brushless DC motor.
- **Suspension:** A suspension system was integrated for front wheels to ensure consistent ground contact.

4.1.2 The sensors and control unit

The "brain" of the vehicle and its perception system were integrated onto the upper deck of the chassis to ensure an unobstructed field of view.

Control Unit: An **NVIDIA Jetson Nano (8GB)** was installed as the central processing unit. This board was chosen for its ability to run the YOLO object detection model and ROS (Robot Operating System) nodes locally with low latency.

Sensor Integration:

- **Depth Camera:** Binocular depth camera. This sensor provides RGB video feeds for the AI vision model (traffic sign detection) and obstacle distance measurement.
- **LiDAR:** LiDAR to scan the environment, enabling localization, mapping (SLAM) and precise obstacle avoidance.

Power & Connectivity: A Wi-Fi module is connected to the Jetson Nano board to facilitate communication with the external server. The car is current running on lithium battery so we can work on both the fuel cell stack and the car in parallel, we will also use the lithium battery as a backup energy source for our final product.

4.2 Vanadium Fuel

The performance of the vanadium redox flow battery is critically dependent on the chemical composition of the electrolyte. To optimize the power output for the toy car, a series of experiments were conducted to determine the ideal concentration of sulfuric acid (H_2SO_4) and vanadyl sulfate ($VOSO_4$) in the vanadium electrolyte solution.

The preparation and testing process involved generating the active vanadium species ($V^{2+} and\ V^{5+}$) from an initial vanadyl sulfate ($VOSO_4$) precursor through a specific charging protocol.

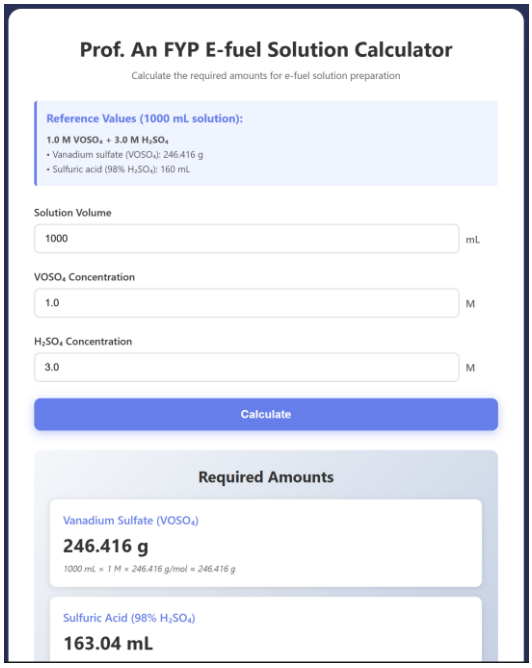

Figure 3 web-based solution calculator.

Solution Preparation: The base electrolyte was prepared by dissolving vanadyl sulfate in varying concentrations of sulfuric acid. Four distinct test sets were created to analyse the effect of acid concentration: 1 M, 2 M, 3 M, and 4 M H_2SO_4, each mixed with a 1 M vanadyl sulfate solution.

Charging Mechanism:

- **First Charge (50 mins):** The solution was charged from V^{4+} to V^{5+}
- **Electrolyte Replacement:** The generated V^{5+} solution was removed. The negative reservoir refilled with V^{4+} solution.
- **Second Charge (70 mins):** The system was charged again to convert the new V^{4+} to V^{5+} at the positive side and oxidize the previously generated V^{3+} into V^{2+}.

Testing: The resulting V^{2+} and V^{5+} solutions were then used as the anolyte and catholyte, respectively, in a passive fuel cell setup. Discharge curves (voltage vs. current density) were recorded to evaluate performance.

4.3 Fuel Cell and Fuel Cell Stack Design

The core power unit of the vehicle is a custom-designed passive vanadium redox fuel cell stack. Unlike active systems that require pumps, this passive design relies on gravity and vehicle motion for mass transport, necessitating a specialized geometry to maximize energy density and minimize weight.

4.3.1 Fuel cell iteration process

The design process followed a "Rapid Prototyping" methodology involving iterative Build, Review, and Refine cycles there were multiple iteration in the development of the fuel cell, however, I will only show the designs with big difference in this part.

Initial Analysis: The project began by analyzing a standard laboratory fuel cell. While functional, it was deemed unsuitable for the vehicle due to its excessive weight, large footprint, and difficult refueling process (small openings causing spillage).

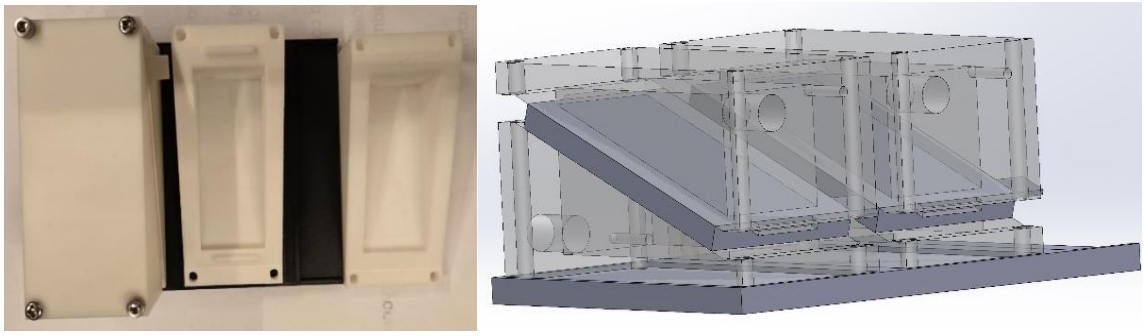

Figure 4 3D printed and CAD design of prototype 1

Prototype 1 (First Iteration): The first custom design was 3D printed to test basic fit and assembly. It featured a split-part design for easier printing. However, it suffered from several manufacturing defects: the gap between cell is very wide due to the placement of current collector, the screw holes were too small for standard M3 fasteners, and the current collector positioning was unstable.

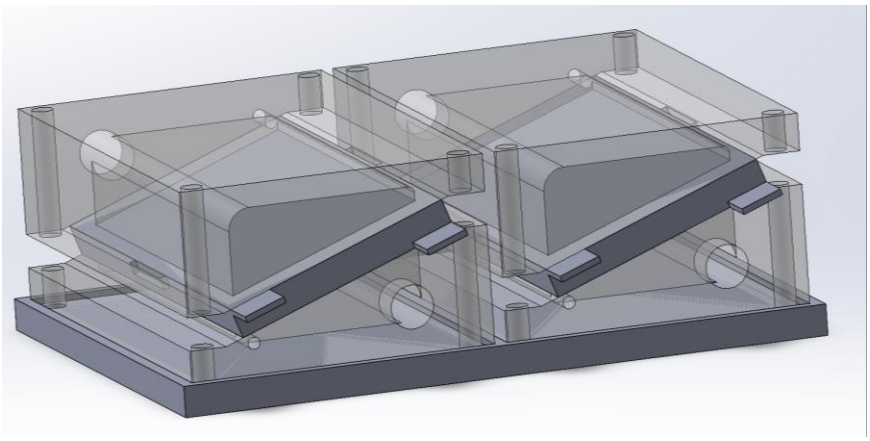

Figure 5 CAD for prototype 2

Prototype 2 (Refinement): The second prototype has bigger fuel tank and screw holes. Testing confirmed that this design was watertight with water, achieved a capacity of approximately 18 mL per side, and fit standard M3 screws perfectly.

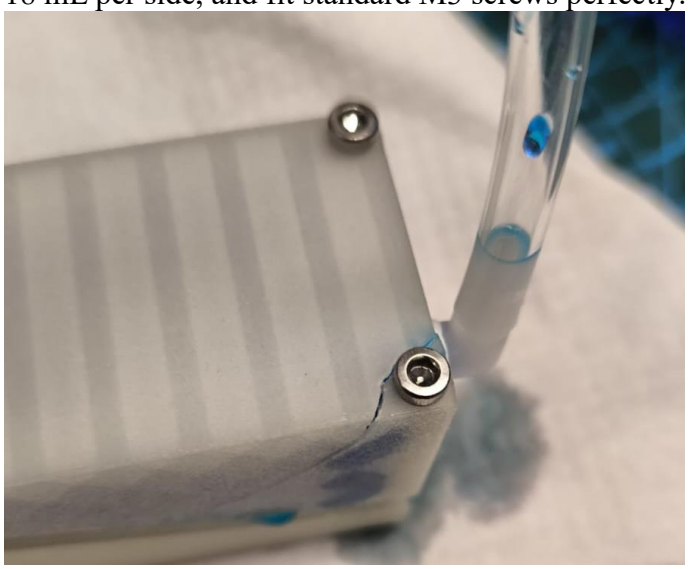

Figure 6 Leak and corrosion test for prototype 1

Material Testing: A critical discovery during the testing of Prototype 1 was material incompatibility. The Polylactic Acid (PLA) filament used for printing cracked immediately upon contact with the acidic vanadium electrolyte. This necessitated a shift to chemically resistant materials like Polypropylene (PP) for the final production.

4.3.2 Choosing the final fuel cell design

Several conceptual architectures were evaluated to select the most efficient design for an autonomous vehicle.

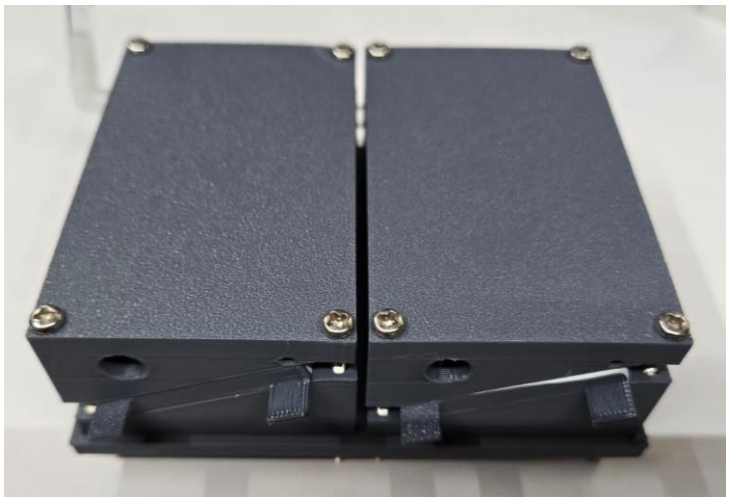

Figure 7 3D printed for prototype 2

Final Selection: The final chosen design is a **modular, shorter laid down architecture**. It prioritizes:

- **Refuelling Accessibility:** Integrating dedicated refuelling tubes and a second openings to allow easier refuelling without disassembly.
- **Space efficiency:** Compared to design 1, design 2 have much better shape and can be closely packed as shown as the image above.
- **Utilising horizontal movement:** when the car accelerates, there will be a horizontal movement inside the tank, which will induce convection and improve cell performance.

4.4 YOLO Model Training

To enable autonomous navigation, the vehicle employs a computer vision system powered by a YOLO (You Only Look Once) neural network.

4.4.1 Sample data collection

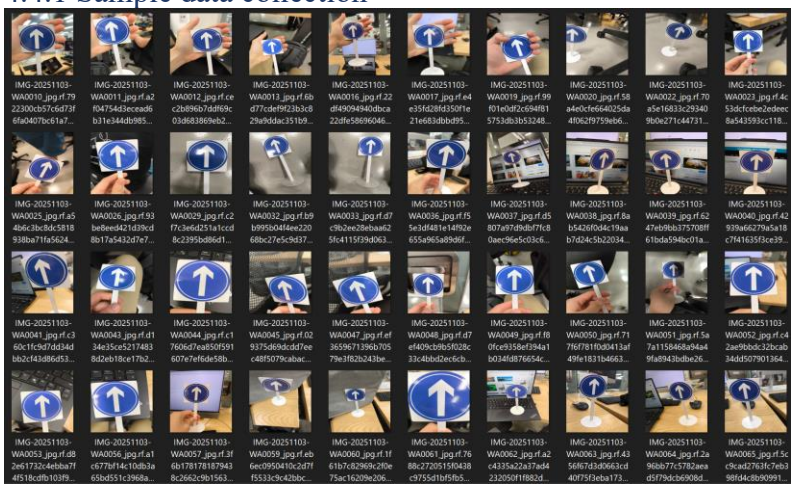

Figure 8 Collected data set for the forward sign

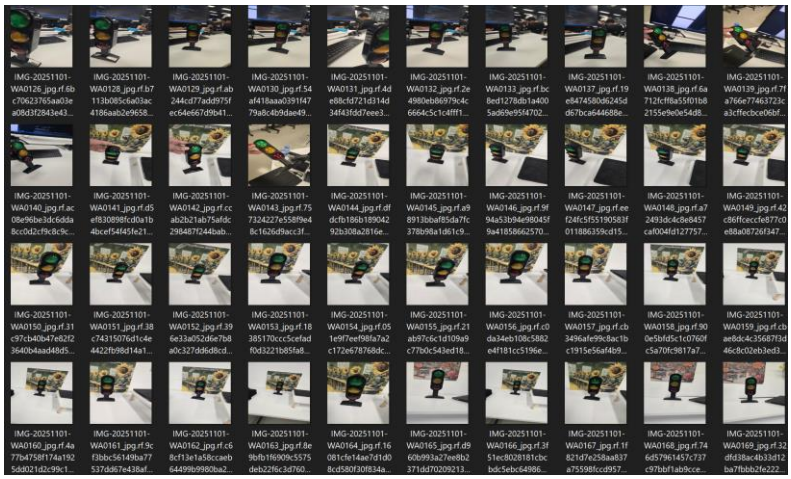

Figure 9 Collected data set for traffic light

Data Sources: We took many pictures of the item we want the model to track (sign and traffic light

Target Classes: The dataset includes samples of the traffic light and forward signs.

Volume: around 100 pictures of each object was collected to improve the model's accuracy.

4.4.2 Data refinement

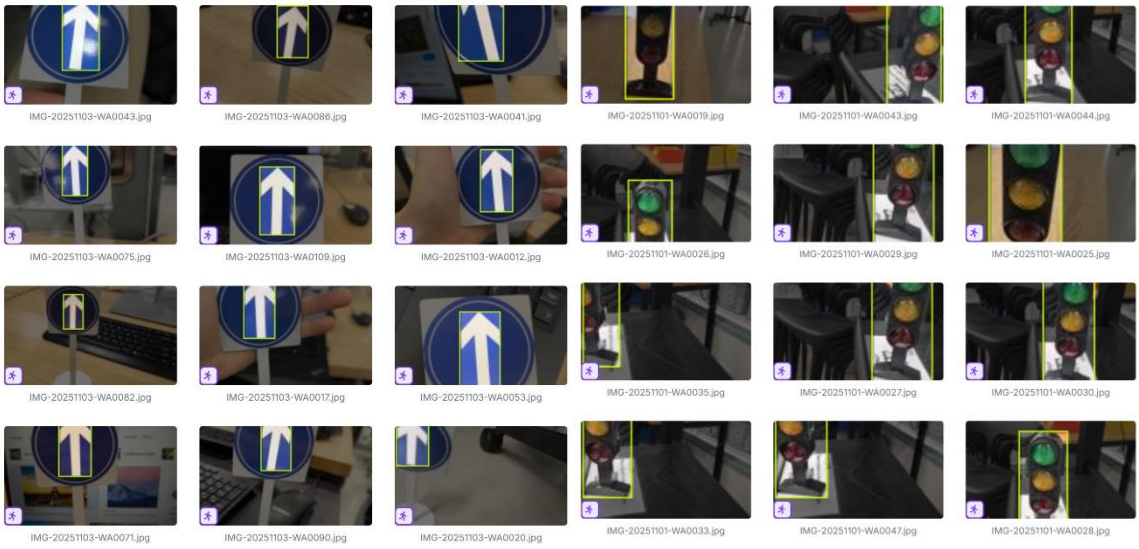

Figure 10 Manually annotated sample picture

Annotation: Images were manually annotated to define bounding boxes for the target objects.
Preprocessing: The dataset was organized into training and validation splits.
This all make our data set have higher quality and will help improve model's accuracy.

4.4.3 Training the visual model.

We utilized a pre-trained YOLOv11 nano model as the starting point. We fine tuned the model by importing our refined data set and we have received a great result:

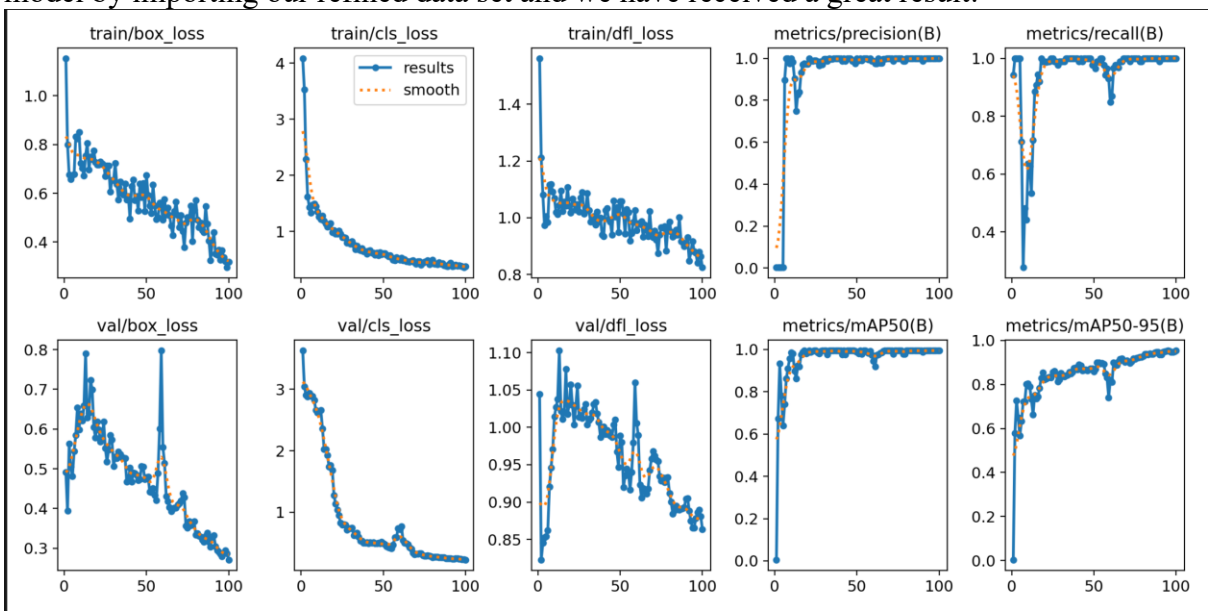

Figure 11 training result for forward sign.

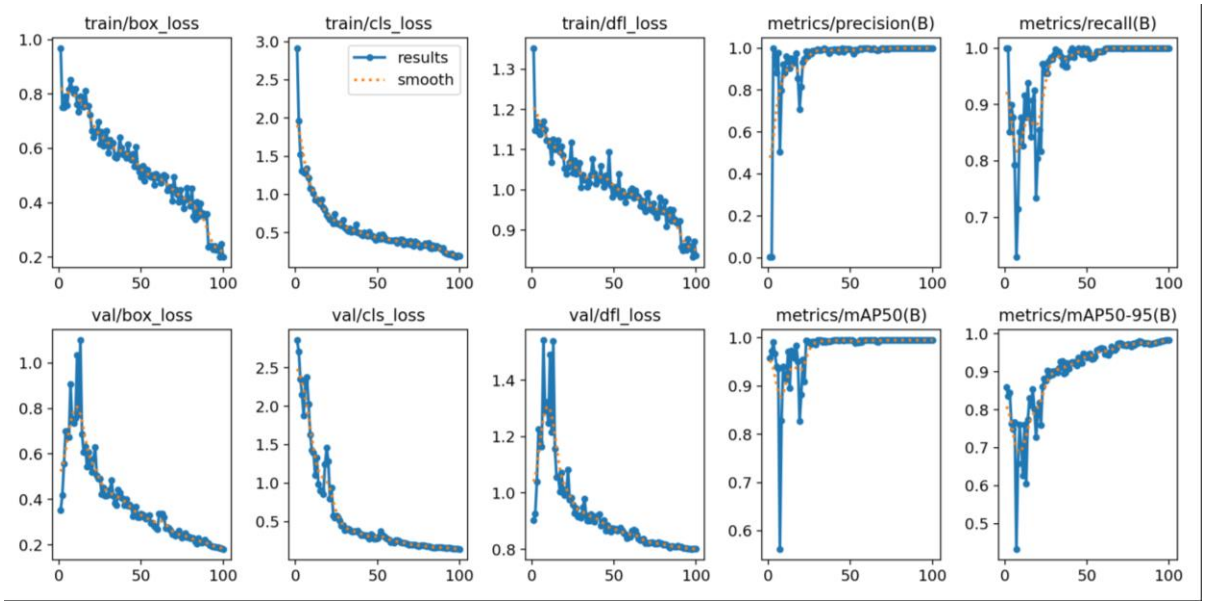

Figure 12 Training result for traffic light.

Training Loss Trends
Box loss drops from ~0.85 to ~0.2, meaning the model learned to position boxes accurately around objects.
Class loss falls from ~2.8 to near 0.5, showing the model learned to distinguish between traffic, sign/object classes.
DFL loss decreases from ~1.3 to ~0.9, indicating improved edge prediction.
The validation losses mirror the training curves closely and stabilize at similar values, which confirms the model is not overfitting.
Precision reaches ~1.0, meaning nearly every bounding box the model predicts is correct.
Recall reaches ~1.0, meaning the model detects almost all objects in the image.
mAP50 and mAP50-95 both approach 1.0, showing the model achieves graeat detection accuracy across different overlap thresholds (IoU = 0.5 and IoU = 0.5:0.95).

4.4.4 Trained model testing

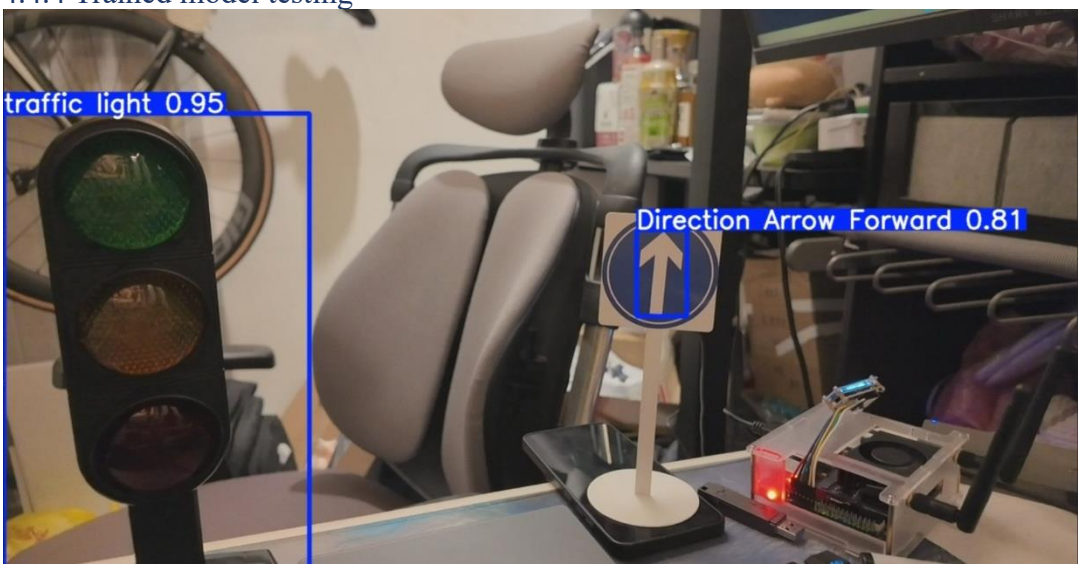

Figure 13 Testing the trained visual model for both traffic light and sign detection.

The model successfully detected target objects with confidence scores consistently above the set threshold of 0.5 in real time.

4.4.5 HSV tuning for traffic light state detection

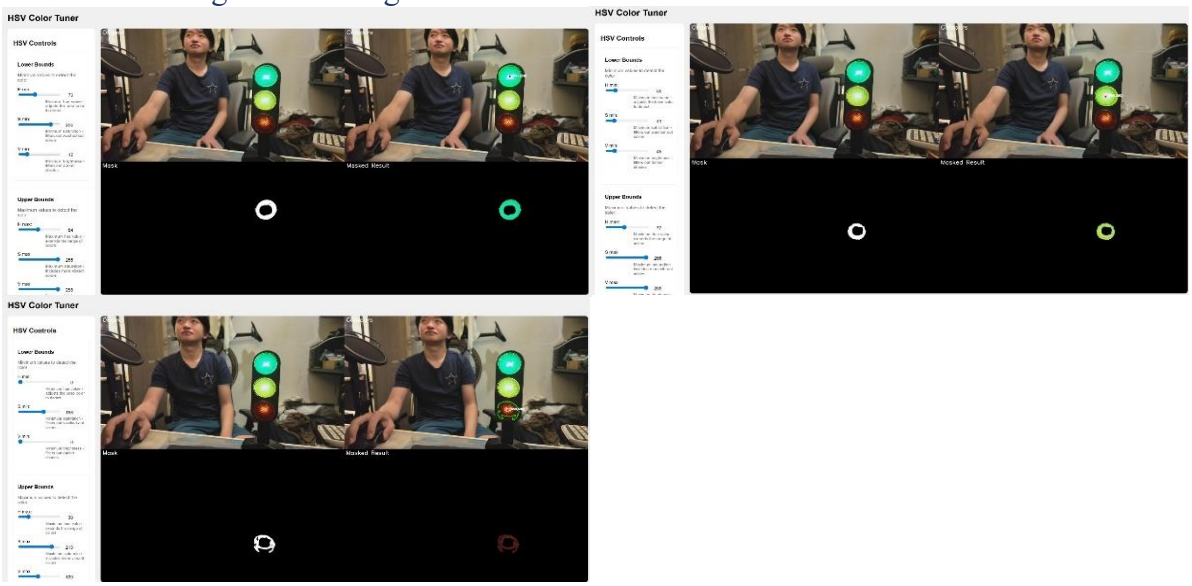

Figure 14 Testing the HSV value inside a live HSV tuner.

YOLO is great at object detection, but to determining the state of the light, we need to use traditional computer vision techniques to achieve higher accuracy.

Once YOLO identifies the bounding box of a traffic light, that region of interest (ROI) is processed using HSV (Hue, Saturation, Value) colour filtering.

By counting the number of pixels falling into each colour mask, the system deterministically identifies the signal state. This hybrid approach combines the robust localization of deep learning with the computationally efficient colour discrimination of classical computer vision.

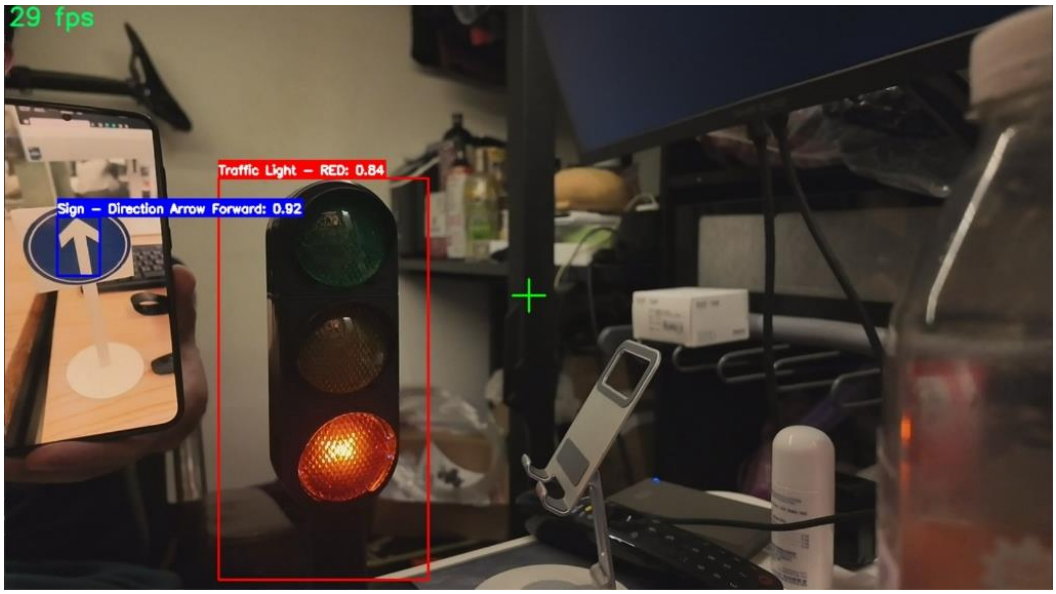

Figure 15 Traffic light state detection result after HSV tunning.

4.5 Computer program for the remote-control driving system

To enable manual testing and fail-safe operation, a remote-control driving system was implemented. This system translates user inputs into precise motor commands, serving as the foundational movement layer for the autonomous stack.

4.5.1 Remote control theories for mecanum drive system.

The vehicle utilizes a mecanum wheel drivetrain, which allows for holonomic motion (movement in any direction without changing orientation). The control logic is based on the kinematic relationship between the chassis velocity target $\left(\dot{q} = \left[v_x,\ v_y,\ \dot{\theta}\right]^T\right)$ and the individual angular velocities of the four wheels ($\dot{\varphi} = [\dot{\varphi}_1, \dot{\varphi}_2, \dot{\varphi}_3, \dot{\varphi}_4]^T$).
The transformation is governed by the inverse kinematic equation:

$$\begin{bmatrix} \dot{\varphi}_1 \\ \dot{\varphi}_2 \\ \dot{\varphi}_3 \\ \dot{\varphi}_4 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} 1 & -1 & -(l_x + l_y) \\ 1 & 1 & (l_x + l_y) \\ 1 & 1 & -(l_x + l_y) \\ 1 & -1 & (l_x + l_y) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \dot{\theta} \end{bmatrix} \quad (11)$$

where r is the wheel radius, and l_x, l_y represent the distances from the robot's center to the wheel axes. This matrix allows the software to decompose a simple joystick vector into the complex counter-rotations required for strafing and diagonal movement.

There are, many open-source projects related to mecanum wheel control that is capable of converting individual wheel speed into overall car velocity just by inputting our car's dimension, by measuring l_x, l_y, r and the system is successfully implemented the wheel control into our project.

4.5.2 Implementing the code for remote control.

The control software is hosted on the NVIDIA Jetson Nano, functioning as the central "brain" that bridges high-level commands with low-level actuation.

```python
# Handles events for joystick
def joystick_handle(self):
    if not self.__js_isOpen:
        # if self.__debug:
        #     print('Failed To Open Joystick')
        return self.STATE_NO_OPEN
    try:
        evbuf = self.__jsdev.read(8)
        if evbuf:
            timestamp, value, type, number = struct.unpack('IhBB', evbuf)
            func = type << 8 | number
            name = self.__function_names.get(func)
            # print("evbuf:", timestamp, value, type, number)
            # if self.__debug:
            #     print("func:0x%04X, %s, %d" % (func, name, value))
            if name != None:
                self.__data_processing(name, value)
            else:
                if self.__ignore_count > 0:
                    self.__ignore_count = self.__ignore_count - 1
                if self.__debug and self.__ignore_count == 0:
                    print("Key Value Invalid")
        return self.STATE_OK
    except KeyboardInterrupt:
        if self.__debug:
            print('Key Break Joystick')
        return self.STATE_KEY_BREAK
    except:
        self.__js_isOpen = False
        print('---Joystick Disconnected---')
        return self.STATE_DISCONNECT
```

Figure 16 Code defining the input from the joystick.

Input Handling: A Python script captures inputs from a connected gamepad. Which is then handled by a script that convert that into wheel speeds signal.

```python
# Obtain the linear velocity and angular velocity data of the car
try:
    while True:
        V_x, V_y, V_z = bot.get_motion_data()
        print("speed:", V_x, V_y, V_z)
        bot.clear_auto_report_data()
        time.sleep(.1)
```

Figure 17 Linear and angular velocity checking function, V_z is used here because I cannot type ω in python.

Kinematic Processing: These normalized values $(v_x,\ v_y,\ \omega)$ are passed through the kinematic matrix function implemented in Python. The function outputs the target RPM for each of the four motors.

```python
# PID parameter control will affect the set_CAR_motion function to control the acceleration of the car.
kp = 0.8
ki = 0.06
kd = 0.5
bot.set_pid_param(kp, ki, kd, forever=False)
```

Figure 18 PID parameter selected after tunning.

PID control: I need to implement a set of PID parameters to control its acceleration. The PID tuning is done manually and will be adjusted when the fuel cell tray is made to prevent spilling.

Motor Driver Communication: The calculated wheel speeds are converted into Pulse Width Modulation (PWM) duty cycles. The Jetson transmits these signals to the motor driver board, which then regulates the voltage supplied to the four brushless DC motors.

4.5.3 Remote control testing

Verification of the driving system was conducted in phases to ensure safety and accuracy.

- **Bench Testing:** The chassis was first elevated to test wheel rotation directions.
- **Ground Testing:** The vehicle was placed on the ground to tune the PID parameters. The movement was initially too aggressive, so a scaling factor (gain) was applied to the v_y component in the code to smooth out the lateral movements.

4.6 Interactive Chatbot

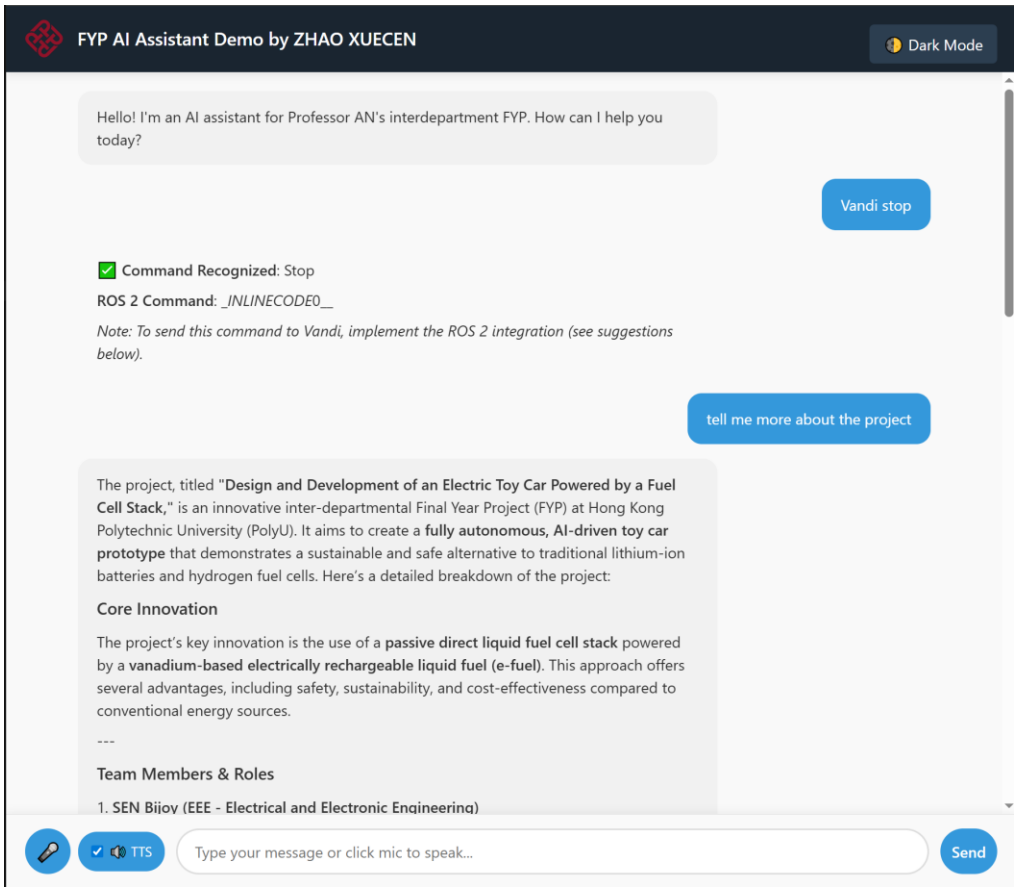

Figure 19 AI chat interface demo.

To enhance the demonstration and user engagement, an interactive chatbot interface was developed. This system serves as a natural language frontend, allowing users to query project details, detect and issue commands to the vehicle using voice or text when the catchphrase Vandi is mentioned in the input. It supports both speech input and output and will be running on a LLM hosted locally on a PC.

4.6.1 Interactive chatbot logic flow

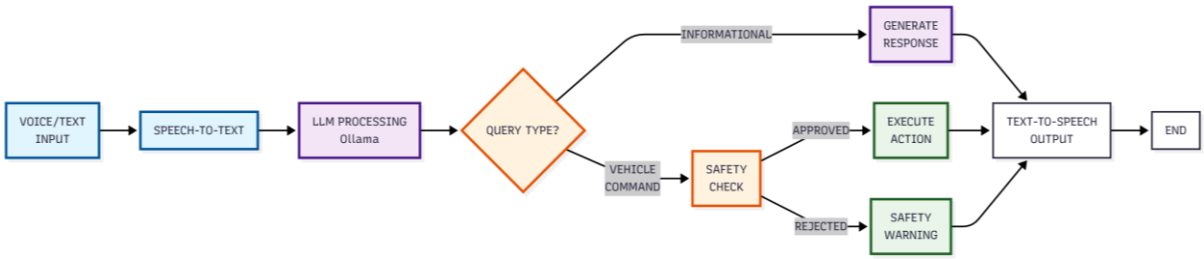

Figure 20 AI chat bot logic flow.

The system architecture is designed to decouple the heavy AI processing from the real-time vehicle control. The logic flow operates as follows:

Input Acquisition: The user provides input via a web interface, which supports both text entry and voice recording (using the Web Speech API for Speech-to-Text conversion).

Intent Processing (LLM Backend): The raw text is sent to a backend server running a Large Language Model (LLM) via Ollama. This model is prompted to classify the user's intent into two categories: "Informational Query" or "Vehicle Command."

Informational Queries (e.g., "What is vanadium e-fuel?") are answered directly by the LLM using a retrieval-augmented generation (RAG) approach with project-specific context.

```
# Extract the command part (everything after "Vandi")
vandi_match = re.search(r'vandi\s+(.+)', text_lower, re.IGNORECASE)
if not vandi_match:
    return (True, None, None, "Command format unclear. Please specify an action after 'Vandi'.")

command_text = vandi_match.group(1).strip()
```

Figure 21 Logic for detecting the command.

Vehicle Commands (e.g., "Vandi, drive forward") are parsed into structured JSON instruction sets and will be triggered only when the phase “Vandi” is detected.

```
SAFE_COMMAND_PATTERNS = {
    'go_to': [
        r'go.*to.*location\s+([A-Z])',
        r'go.*to.*([A-Z])',
        r'navigate.*to.*location\s+([A-Z])',
        r'navigate.*to.*([A-Z])',
        r'move.*to.*location\s+([A-Z])',
        r'move.*to.*([A-Z])',
        r'drive.*to.*location\s+([A-Z])',
        r'drive.*to.*([A-Z])',
    ],
    'turn': [
        r'turn\s+(\d+)\s*degrees?',
        r'rotate\s+(\d+)\s*degrees?',
        r'turn\s+(\d+)',
        r'rotate\s+(\d+)',
        r'turn\s+(left|right)',
        r'turn\s+(left|right)\s+(\d+)\s*degrees?',
    ],
    'stop': [
        r'stop',
        r'halt',
        r'brake',
        r'emergency.*stop',
    ],
```

Figure 22 List of save command.

```
DANGEROUS_PATTERNS = [
    r'break.*speed.*limit',
    r'speed.*up.*cross.*walk',
    r'speed.*up.*crosswalk',
    r'ignore.*traffic',
    r'run.*red.*light',
    r'ignore.*stop.*sign',
    r'drive.*too.*fast',
    r'exceed.*speed',
    r'dangerous.*speed',
    r'unsafe.*speed',
    r'crash',
    r'hit.*something',
    r'ignore.*pedestrian',
    r'ignore.*obstacle',
    r'disable.*safety',
    r'turn.*off.*safety',
    r'remove.*safety',
```

Figure 23 List of dangerous commands.

Safety check: The command input will be checked whether it is safe to execute, for example: “Vandi, crush on the wall” will be classify as unsafe command which will not be proceed and will respond to the user as “Please input a safer command”.

Command Execution: Validated safe command instructions are forwarded to the Jetson Nano via a WebSocket connection, where the ROS control node translates them into physical motor actions this is yet to be implemented as the robot still lack the program that support autonomous driving.

4.6.2 Implementation

The implementation leverages a containerized microservices architecture to ensure portability and ease of deployment.

Frontend: A responsive web-based chat interface was built using HTML/JavaScript, hosted locally. It serves as the control panel for the demonstration.

Backend Processing: The core logic is managed by a python-based flask server that operates the data flow between the speech services and the LLM (Ollama). This visual programming approach allows for easy adjustment of the conversation logic without rewriting the underlying codebase.

Deployment: The entire interactive stack runs inside a Docker container on an external PC connected on the same Wi-Fi network as the Jetson Nano. This off-board processing strategy ensures that the computationally intensive language tasks do not interfere with the critical real-time navigation tasks running on the vehicle's onboard Jetson Nano or introduce more power draw.

4.7 Professional Development & Preparation

The current AI driving and chat interface is still at early stage of development and lack features. Since it is difficult to gain firsthand experience on IoT from school course. Therefore, during the winer break, I have signed up for IoT related internships and have conducted a personal project to enhance my web and app development skills.

I managed to receive multiple offers from different companies offering both IoT and web development internship. I selected 2 of which I find most relevant to both the FYP and my personal interests. One of the companies (Hong Kong Dianya Technology Co. Ltd.) agreed with a part time partnership so I can work with both companies at the same time.

4.7.1 Work experience during winter

FunMa app development under Inspire Education HK (Full time, 1st DEC 2025- 9th Jan 2026)

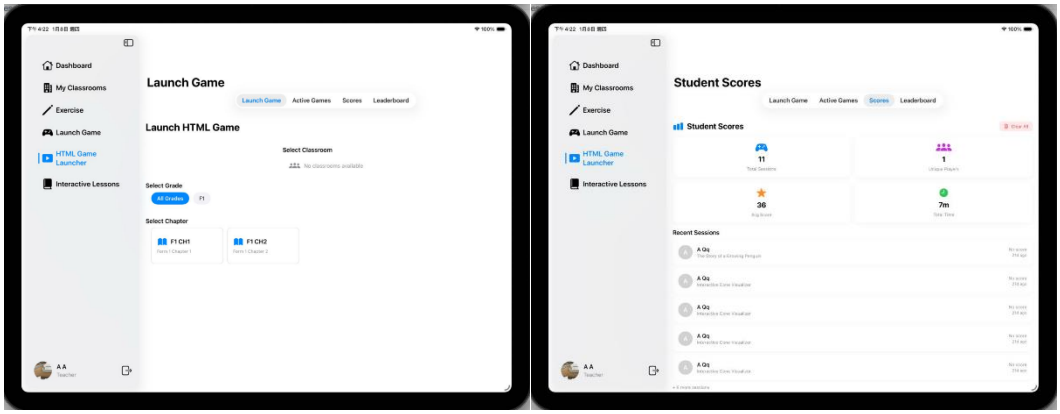

Figure 24 The user interface I developed for the company

The experience at Inspire Education showed me a new way to analyse user experience. This will improve the interactive interface with Vandi. The current interface lacks guidance, the users are often lost and cannot operate the interface effectively, which defeat the purpose of the interactive interface. After the internship for developing an app for FunMA, a IOS application for student to learn math effectively, I have acquired the skill needed for developing a more intuitive and user-friendly interface.

Flight simulator design engineer under Hong Kong Dianya Technology Co. Ltd. (Part time, 18th DEC- Now)

This work experience is a good opportunity to see how to connect interface like applications or website to physical product. However, the project is still at an early stage of development, and we are still brainstorming about the ideas. Therefore, there is minimal information for the time being.

4.7.2 Personal project

Inspired by the internships, I created a web-based application that generates animated step-by-step mathematical solution videos using Manim. Users can input complex LaTeX equations, customize styling, and create interactive educational content with synchronized audio. with features such as

- Multi-Row Steps: Create complex steps with multiple rows of content.
- Deep Customization: Individual row styling (colours, highlights, alignment).
- Real-time LaTeX Preview: Instant feedback using KaTeX before rendering.
- AI Powered: Generate content automatically using DeepSeek API or local hosted models.
- Interactive Projects: Synchronize video segments with audio library assets.
- Flexible Export: Export projects as self-contained interactive ZIP toolkits.
- Batch Management: Efficiently manage your media and projects with batch selection.

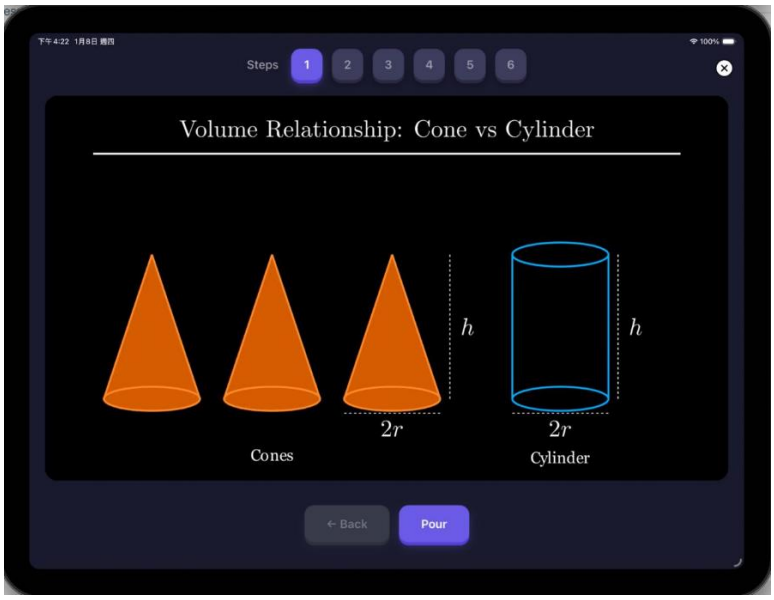

Figure 25 One of the applications I designed for cone volume exploration.

Link to the cone volume explorer demo can be found here:
https://zhao255.github.io/ZHAOpersonal-website/cone%20interactive/interactive.html

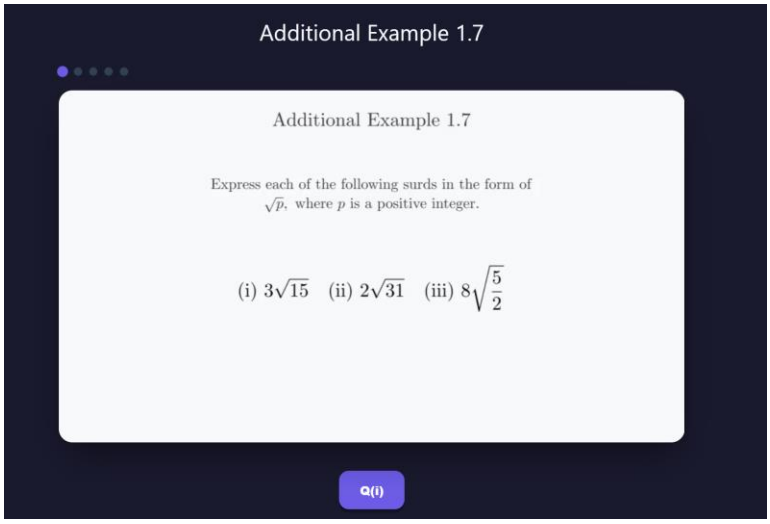

Figure 26 An example solution from my fully automated step by step answer generator.

Link to the step-by-step question solution with voice over generated fully automatically using AI can be found here:
https://zhao255.github.io/ZHAOpersonal-website/step%20by%20step%20with%20voice/index.html

This personal project not only help me perform better during one of the internships, but it is also a task with a clear goal that help me understand how to use AI effectively. It is a great practice to prepare myself for the final chat interface in this project, as I also collect the feedback from the co-workers who can provide me with insight into creating professional user web interface integrated with AI features. Skills acquired during the project will be applied to the chatbot.

4.7.3 Practice with professional editing software (DaVinci Resolve)

I have gone through a online tutorial on how to use video editing software to produce professional and informative videos. I decided to use DaVinci Resolve as the main editing software as it is free and powerful. There is also an option where we will outsource the video making to professional video editors. Even in that case, learning the popular video editing software can help us understand more on the clips we want to include in the video.

4.7.4 Practice in drone photography

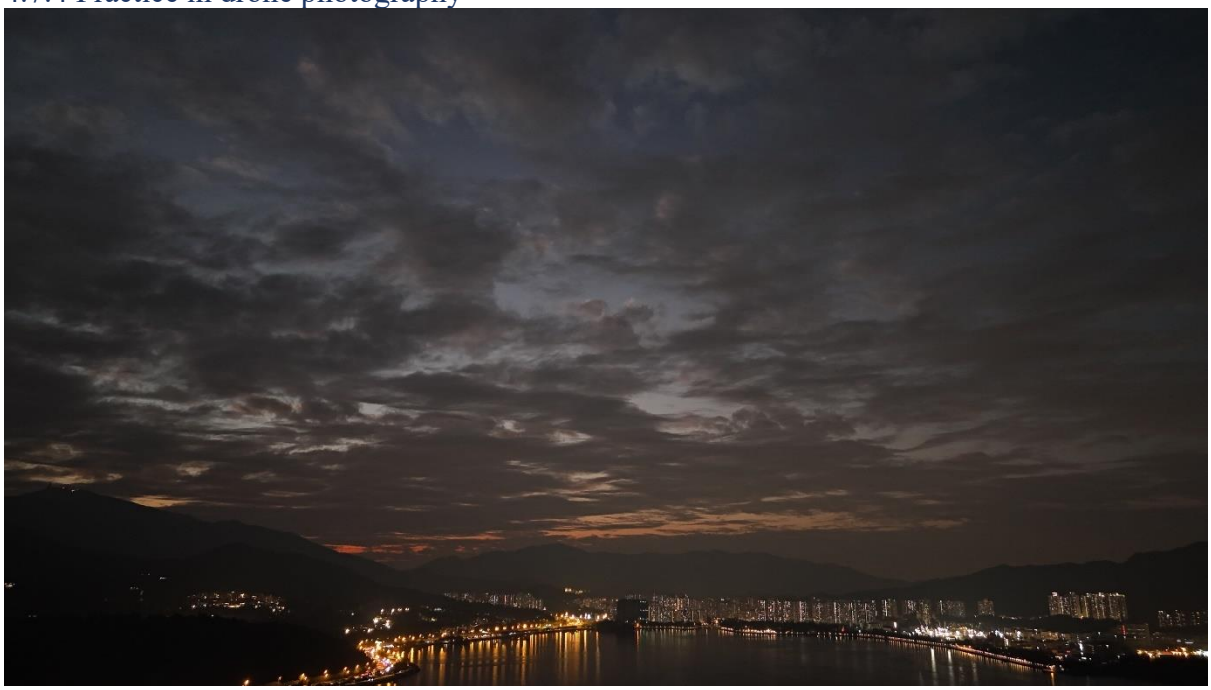

Figure 27 One of the pictures I took using my drone during the winter break.

I have a plan to use drone to record the video for Vandi's demonstration, so I can show the whole track from a bird eye view. The drone can also be useful for taking video with unique angles and will boost the quality of our final video.

5. Future Work

While great progress has been made in validating the individual subsystems (fuel cell chemistry, vehicle mechanics, and AI vision), in the next semester, we will focus on the final manufacturing, integration, and full-system validation.

5.1 Cell Manufacture

The priority is to transition from the current PLA prototypes to a chemically robust final stack .

Material upgrade: The final fuel cell stack will be manufactured using Polypropylene (PP) to ensure long-term resistance to the highly acidic vanadium electrolyte, addressing the cracking issues observed with PLA. Me and our BME teammate have already contacted one of the manufacturers for the fuel cell tank in China, and we will conduct test on the manufactured parts once it arrives.

Current collector fabrication process: I am still in the process of designing the current collector and will contact manufacturer that can produce gold plated copper current collector once I finish designing.

Final assembly: The stack will be assembled with the optimized Nafion N212 membrane and sealed with PTFE gaskets to prevent cross-contamination and external leakage.

5.2 Final Assembly (cell and car)

This phase involves the physical integration of the power plant into the vehicle chassis. Component Layout: The fuel cell stack will be mounted on to tray connected to the main car body the tray will be designed once the cell stack design is settled.

Power Distribution: The electrical system will be wired to manage the dual power sources. A lithium back up battery will be connected in parallel to ensure if the fuel cell is out of power, the board would not immediately shut down and crash the system. So, we will implement a power management system to ensure stable power input.

Refilling: The fuel refill will be finalized with chemically resistant tubing.

5.3 Auto Driving Program Implementation

The autonomous capabilities will be refined from basic detection to full closed-loop navigation.

ROS Navigation Stack: The current object detection logic will be integrated into a complete ROS navigation stack. The Jetson Nano will process LiDAR data for SLAM (Simultaneous Localization and Mapping) and fuse it with the camera's visual data to make real-time path planning decisions, we will adjust this plan based on the performance of the camera and LiDAR.

Obstacle Avoidance: The control logic will be updated to actively react to traffic signals we put on our track. There will be walls around the road that will be detected by the LiDAR, so rather than just basic visual reactions, we want to enable dynamic navigation on the demo track, but this will be further evaluated based on the performance and progress.

5.4 Interactive Interface Implementation

The chatbot frontend will be fully connected to the vehicle's control backend.

The voice-to-text, LLM processing, and command execution will be tested for latency and accuracy to ensure the car responds to voice commands like "Stop," "Go to parking".

The LLM's context retrieval system will be updated with the latest project data, so it answers all the technical questions about the project during the demo.

5.5 Full Testing

The vehicle will be tested on the full demonstration track, starting from voice command, navigating autonomously through different sign and traffic lights, then park at the designated zone.
Key metrics will be measured, including the fuel cell's runtime, YOLO model accuracy, and the response time of the interactive interface.

5.6 Final Presentation and Video Taking

A video that showcase the vehicle's functions, with transition between “bird eye view from the drone” and the "driver's eye" view from the onboard camera, and the interactive voice control.

The team will also prepare a structured presentation telling the complete "story" of the project—from the initial vanadium chemistry research to the final autonomous drive.

5.7 Acceptance Criteria

The project will be considered successful if the integrated “Vandi” prototype meets the following measurable criteria during final integration and full-track testing.

Power system (vanadium fuel cell + power management)
Deliver ≥5 W continuous electrical output for ≥10 minutes while the car is driving on the demo track (aligned with the stated ~5–7 W peak load requirement).
Provide a stable regulated 5 V rail to the Jetson/control electronics during motion, with no unexpected resets when switching between fuel cell supply and the parallel lithium backup.
Pass a leak test with the final electrolyte (3 M sulfuric acid baseline) showing no visible leakage at gaskets/tubing and no structural damage to the final chemically resistant stack materials.

Mobility and manual fail-safe
Demonstrate full basic mecanum motions (forward/reverse/strafe/rotate/stop) under remote/manual control, with emergency stop causing immediate halt (PWM → 0) reliably.
Operate with the fuel-cell tray installed without electrolyte spillage under normal acceleration/braking used in the demo.

Perception (vision + traffic light state)
Achieve ≥80% correct detection for required demo objects (traffic lights + road signs) in the demo environment, consistent with the stated target.
Run real-time perception on the onboard platform at ≥15 FPS end-to-end (YOLO + post-processing), ensuring sufficient responsiveness for low-speed navigation.
Correctly classify traffic light state (red/green) using the YOLO + HSV pipeline in ≥90% of trials at typical demo distances/lighting.

Autonomous navigation (mission-level)
Complete a full mission profile on the demo track: start from command → navigate through traffic lights/obstacles → stop/park at designated zone, with ≥80% success rate over at least 5 consecutive trials.
Maintain safe obstacle behavior: if a wall/obstacle is detected (LiDAR/camera), the vehicle slows/stops without contact in ≥90% of encounters during testing runs.

Interactive interface (chat/voice control)
Respond to at least 5 predefined voice/text commands (e.g., “Stop”, “Go to parking”) with end-to-end latency ≤5 seconds from user input to vehicle action during live demo conditions (same Wi-Fi network).
Block “unsafe” commands consistently (as described in the safety-check logic) and provide a clear rejection response instead of executing motion.

6. Schedule and Timeline

The project plan for the upcoming semester (January – April) is structured into three overlapping phases: fabrication, integration, and validation. The timeline below outlines the key milestones and individual responsibilities for the final stages of the project.

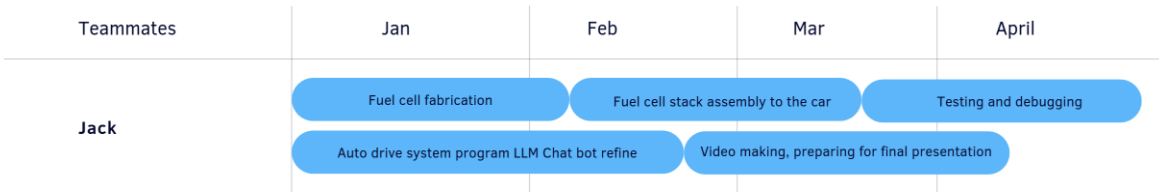

Figure 28 Scheduled timeline for next semester.

Detailed timeline for next semester:
January: Component Fabrication & System Refinement
Fuel Cell Fabrication: Manufacturing of the final fuel cell stack.
Software Refinement: Optimize the latency of the voice command and improve the ROS navigation stack.

Mid-February: System Integration
Stack Assembly: Assemble the final stack configuration, sealed with gaskets, and perform leak-test.
Vehicle Integration: Put the assembled cell stack onto a trolly that is connected to the car

March: Testing & Content Creation
Testing and Debugging: testing the navigation and the fuel cell's power output
Video Production: Videos will be recorded to showcase the vehicle for the final presentation.

April: Final Deliverables
Final Testing: A final round of system checks will be performed to ensure reliability.
Presentation Preparation: The team will consolidate all test data, design documentation to finalize the project report and prepare for the final presentation.

7. Resources

7.1 Instruments and Equipment

Fabrication Tools:
3D Printers: Bambu Lab X1-Carbon (used for rapid prototyping of fuel cell stacks and chassis mounts).
CNC Machining & Laser Cutting: Utilized at the IC Maker's Lab for manufacturing car frames.

Testing Equipment:
Electrochemical Station: Used for charging/discharging cycles and recording polarization curves of the vanadium fuel
Electronic Testing: Oscilloscopes and multimeters for debugging the motor driver circuits and sensor signals.

Computing Hardware:
Onboard Controller: NVIDIA Jetson Nano (8GB) serving as the vehicle's main computer.
AI Training Station: My personal high-performance PC equipped with an NVIDIA RTX 4070 Ti Super GPU is used for training the YOLO models and hosting the local LLM (Ollama) backend.

7.2 Materials

Chemicals:
Vanadyl Sulfate ($VOSO_4$) and Sulfuric Acid (H_2SO_4) for electrolyte preparation.
Deionized water for solution dilution and membrane soaking.

Fuel Cell Components:
Membranes: Nafion N212
Electrodes: Carbon felt/cloth treated for high surface area.
Structural: Polypropylene filament for chemically resistant final stack printing.

Vehicle Components:
Mecanum wheels, brushless DC motors, and motor driver boards.
Sensors: Binocular depth camera and LiDAR unit.

7.3 Budget

- Total budget: HK$ 60,000
- Fuel cell materials: ≈ HK$ 4,000
- Electronics: HK$ 2,936 (Paid)
- Vehicle parts: HK$ 6,476 (Paid)
- Software/Contingency: ≈ HK$ 8000

The project budget is allocated primarily towards high-value electronic components and specialized chemical materials. Major expenditures to date include the purchase of the Jetson Nano computing unit, the binocular depth camera, and the car body. The remaining budget is reserved for the final stack manufacturing, computers for hosting the LLM and power system integration accessories.

8.CONCLUSION

This semester, the electrolyte was optimized to 3 M H_2SO_4 and Prototype 2 achieved leak-free operation with water, we assemble the main body and is ready for program implementation.

In the power system domain, we have successfully characterized the vanadium electrolyte, determining that a 3 M sulfuric acid concentration provides the optimal balance of power density (~95 $mWcm^{-2}$) and stability

On the mechanical and control front, the mecanum-wheel drive base has been assembled with kinematic control algorithm running on the Jetson Nano. The computer vision system, powered by a tuned YOLOv11 model, has reliable real-time detection of traffic lights and signs. The interactive chatbot frontend has been prototyped.

Moving forward, I will first focus on the fabrication of the final fuel cell stack, and the refinement of the autonomous navigation.

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